

Project Logbook — Personalized Weather Station

Phase 1 — Project Ideation & Planning (Late January – Early February)

- Project initiated in late January 2026
- Defined project goals: real-time environmental monitoring of temperature, humidity, and atmospheric pressure
- Selected BME280 as primary sensor for its multi-parameter capability
- Initial system concept built around Raspberry Pi 4 as the processing platform
- Established GitHub repository for version control and documentation
- Created initial project definition document outlining objectives, constraints, and system components

Phase 2 — Proof of Concept on Raspberry Pi (February – Early March)

- Began sensor integration on Raspberry Pi using breadboard
- BME280 units received but immediately exhibited problems — incorrect pin labeling caused improper connections, damaging the sensors during initial testing
- I2C communication could not be stabilized despite multiple wiring configurations and library implementations
- Decision made to pivot to DHT22 as a temporary replacement — already on hand, functioned reliably, and validated the core sensing approach
- DHT22 successfully integrated with GPIO4 on the Raspberry Pi at 3.3V logic
- Implemented retry logic (3 attempts per cycle) to handle timing jitter from Linux's non-real-time OS
- Established 2-second sampling interval matching DHT22 minimum recommended read interval
- Developed Tkinter GUI for real-time display of temperature and humidity readings
- Iterated through 4 software versions — basic read, retry logic, stable console monitor, GUI integration
- System validated for stable temperature and humidity monitoring

Phase 3 — Platform Pivot to Custom PCB (March 5th)

- Professor recommended moving away from Raspberry Pi and touchscreen — impractical for a real manufacturable product
- Team agreed the custom PCB approach was more aligned with course goals and real engineering practice
- Decision made to design a custom PCB around the ATmega328P-A microcontroller
- Replaced Tkinter GUI concept with phone notifications via ntfy for a more practical and scalable user interface
- DHT22 officially dropped in favor of BME280 once a replacement unit was correctly wired and confirmed operational

Phase 4 — PCB Design & Schematic Capture (March – April)

- Schematic designed in KiCad using ATmega328P-A as core MCU and BME280 as primary sensor
- Selected SPI communication based on available KiCad BME280 footprint — device operates over I2C with CSB pin tied high

- External crystal oscillator added per professor recommendation for more precise and stable MCU timing versus internal RC oscillator
- Passive component values selected from datasheets — 4.7k Ω pull-ups, 10k Ω reset pull-up, 330 Ω LED resistor, 22pF crystal load caps
- Decoupling capacitors added on VCC, AVCC, and BME280 VDDIO for power supply noise filtering
- ISP header (J2) included for direct firmware programming onto ATmega328P
- Test points added on PD0, PD1, PD2, and GND for probing and future sensor debugging
- Status LED added on PB4 with current-limiting resistor

Phase 5 — PCB Layout & Design Review (April 1st)

- Partner completed initial PCB layout and sent for design review on April 1st
- Design review identified missing decoupling and bulk capacitors — added during review
- Routing on front and back copper layers created conflicts — limited space for additional traces
- Decision made to experiment with 4-layer stackup to resolve routing constraints and explore inner copper layer routing
- Successfully learned via-to-layer assignment and signal routing across inner copper layers
- 4-layer board adopted as the final submitted design
- ERC and DRC run — all violations resolved
- 3D render generated confirming component placement and board layout

Phase 6 — Manufacturing Preparation (Late April – Early May)

- Gerber files generated for all layers
- Drill file exported
- BOM compiled with all component part numbers and values
- PCB ordering instructions documented — board confirmed manufacturable at JLCPCB, PCBWay, or OSHPark
- Repository organized with all fabrication files, documentation, and design files

Phase 7 — Final Additions & Presentation Prep (May 7th)

- Push button added to schematic today for user-triggered notification functionality
- Phone notification system via ntfy planned for software implementation
- Final presentation prepared covering full design cycle from concept to manufacturable PCB
- Project submitted May 7th 2026

Lessons Learned

- Hardware reliability must be verified before full integration — the BME280 pin labeling issue cost significant time early on
- Timing-sensitive protocols require retry strategies under Linux — not all platforms are real-time
- Design reviews add real value — missing components were caught and added before the layout was finalized
- A 4-layer board opens up significantly more routing flexibility and is worth learning early
- This project had direct real-world application — the skills learned were immediately applied professionally to recreate a discontinued kit from an existing schematic

